

Kyle Main

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Machine Learning Engineer — Detection & Behavioral Analytics

Machine learning engineer and data scientist with 12 years of experience building systems that tell malicious behavior apart from normal behavior at scale — including 8+ years of hands-on ML detection engineering (unsupervised clustering models, time-series anomaly detection, and precision-tuned feature engineering) across massive telemetry volumes. Full-lifecycle ML experience: exploratory data analysis at scale (PySpark/SparkSQL on GCP Dataproc), model development with pandas/scikit-learn/PyTorch, and production deployment of detection content with continuous false-negative/false-positive analysis feeding back into feature engineering — the same efficacy loop central to this role.

CORE SKILLS

ML & Data Science: Python, SQL, pandas, scikit-learn, NumPy, SciPy, PyTorch, clustering/unsupervised ML, time-series anomaly detection, R

Data Analytics at Scale: PySpark / SparkSQL, GCP Dataproc, Zeppelin notebooks, BigQuery, Apache Beam / GCP Dataflow

Detection & Feature Engineering: Signature, statistical, behavioral, and ML-based detection rule development; entity/behavior baseline modeling; false-negative/false-positive analysis and tuning across attack categories

Production ML Pipelines: Detection-as-code CI/CD (GitLab), automated testing, rule-quality metrics tracking, staged/safe production rollout

Engineering: Git, Docker, structured/tested Python code, Elasticsearch

PROFESSIONAL EXPERIENCE

Senior Cybersecurity Engineer — Shorepoint

Oct 2023 – Present

- Built a UEBA (User and Entity Behavior Analytics) detection layer on custom Elasticsearch data transforms, modeling enterprise-wide behavioral baselines from CrowdStrike, Suricata, and Zeek telemetry to surface deviations from normal activity (DOE/NNSA Security Data Integration project, completed).
- Currently build and tune production detection analytics for Treasury's SOC, translating false-positive/false-negative investigation findings directly into refined detection logic (current project, Treasury SOC).

Senior Threat Detection Engineer and Data Scientist — Forescout

Aug 2022 – Oct 2023

- Senior member of a data science and detection engineering team building signature, behavioral, statistical, time-series, and ML-based detection content against massive-scale customer data on a cloud-based big-data platform.

Threat Detection Engineer and Data Scientist — Trend Micro / Cysiv

Sep 2018 – Aug 2022

- Built ML-based detection content clustering devices on the network by behavioral features (unsupervised learning) to surface anomalies signature-based rules missed — part of 2,300+ detection rules covering the MITRE ATT&CK matrix.
- Designed time-series anomaly detection models for entity behaviors — Outlook process chains, parent/child process combinations, and authentication patterns (volume and geography) — to flag deviation from an established baseline, directly analogous to modeling communication-pattern baselines and combining discriminative signals into a precise detection system.
- Ran exploratory data analysis at scale on GCP Dataproc compute clusters via Zeppelin notebooks and PySpark/SparkSQL, building a reusable analysis toolkit to validate new detection signal candidates before productionizing them.
- Owned data engineering feeding the detection pipeline — 220+ ingested sources, 50+ Logstash filters, and a Common Information Model standardizing fields across all of it — so model inputs stayed reliable at production scale.

Security Data Scientist — Experian

Jan 2015 – Jan 2018

- Analyzed large-scale security datasets to build custom models and algorithms identifying emerging threats — DNS-based malware detection/mitigation and anomalous-behavior discovery in log data.

EDUCATION & CERTIFICATIONS

M.S. Physics — University of North Texas (2013) | B.S. Physics — Ball State University (2011)

Splunk User Certification · Splunk for Analytics and Data Science · Johns Hopkins (Coursera): R Programming, The Data Scientist's Toolbox